

EEG Motor Imagery Classification under Unified Evaluation Protocols

J Y · BSc Statistics and Data Science (double major: Computer Science), Korea National Open University

KNOU Statistics & Data Analysis Competition entry (Bronze Prize, 2025), refactored and re-analysed in 2026 · github.com/J-Y00N/EEG-Motor-Imagery-Classification

Introduction

Motor-imagery decoders are often ranked under a single protocol. This project compares classical, geometric and deep decoders on BNCI 2014-001 under separated protocols and an analysis plan fixed before the runs. The 2026 refactor corrected design errors of the 2025 version (a feature-scaling bug, a biased ERD/ERS normalisation, pseudo-replicated statistics) and added the dataset's standard cross-session protocol.

Data and Preprocessing

Data. BNCI 2014-001 (BCI Competition IV 2a): 9 subjects, 2 sessions on different days, left- versus right-hand imagery, 288 trials per subject.

One canonical path. 50 Hz notch, 8–32 Hz band-pass, 0–4 s after the cue; ICA and re-referencing off.

No leakage. Feature selection, scaling and early stopping use training data only.

Subject-level statistics. Mean $\pm$ SD over subjects, exact paired sign-flip tests, Holm correction; three comparisons fixed in an analysis plan before the runs.

Models and Protocols

Models. Raw band power, CSP and filter-bank CSP (FBCSP) with LDA; Riemannian tangent space with LDA; EEGNet (up to 300 epochs, early stopping, mean of 5 training seeds).

Cross-session. Train on session 1, test on session 2 of the same subject (144 trials each).

LOSO. Train on 8 subjects (2,304 trials); matched LOSO uses only 144 of them, tested on the same session-2 trials.

Transfer. Zero-shot and 5–30 calibration trials per class from the new subject.

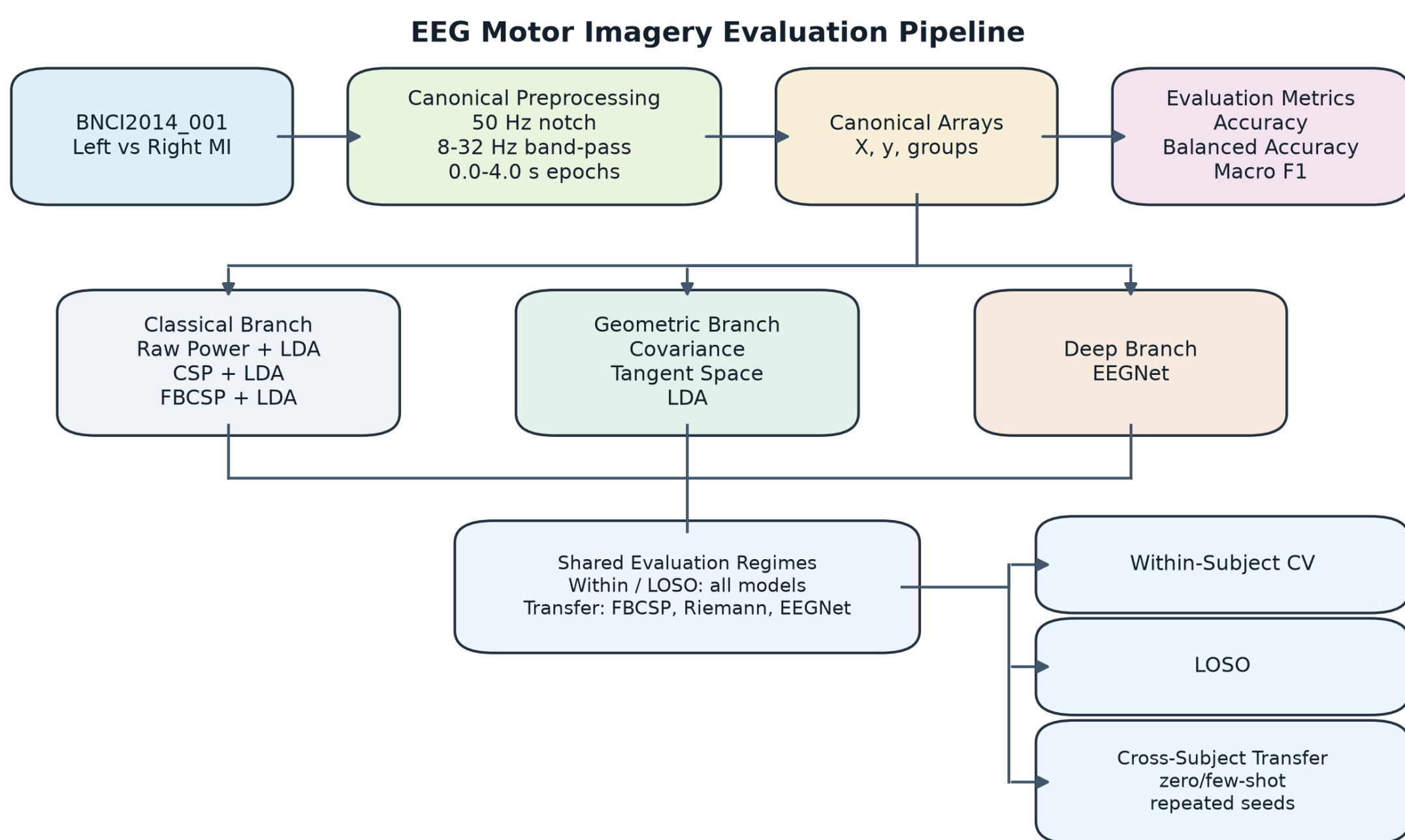


Figure 1. One preprocessing path feeds three model families; transfer uses FBCSP, Riemann and EEGNet.

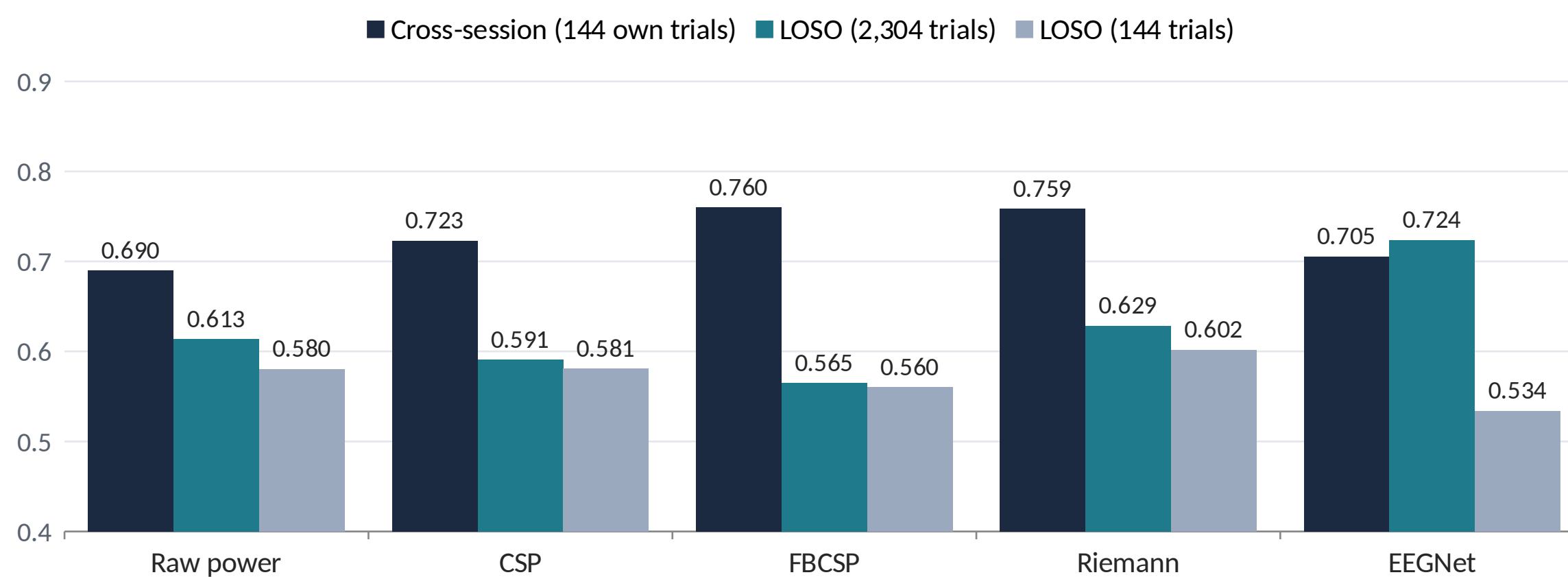
Ranking Follows the Training Data

−0.055

EEGNet – FBCSP · cross-session · n.s.

+0.095

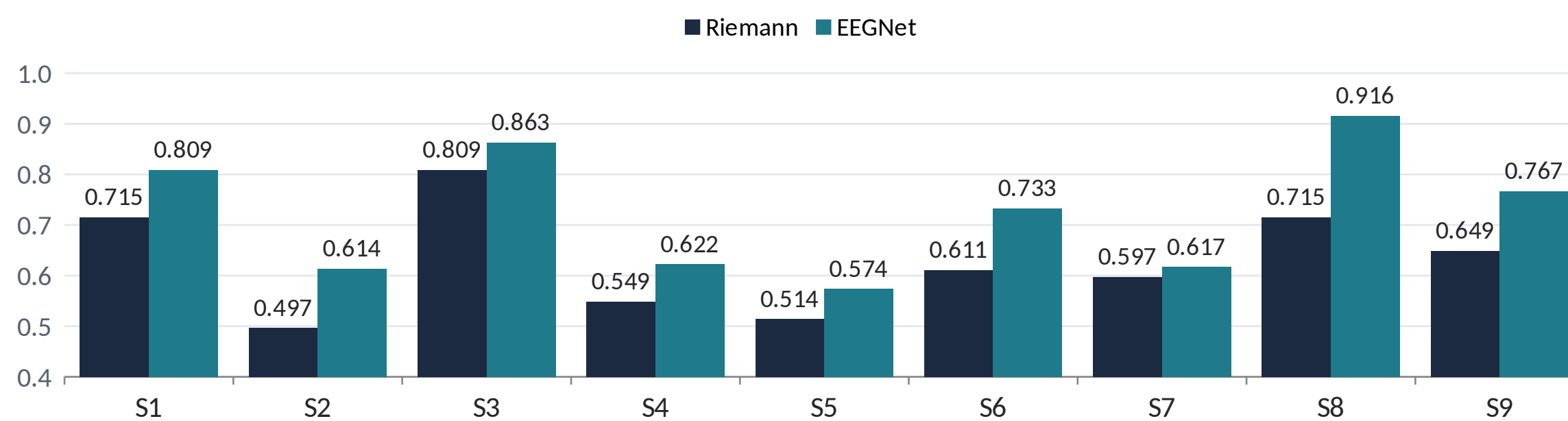
EEGNet – Riemann · LOSO · p = 0.012



EEGNet leads only with 2,304 pooled training trials. With 144 trials from other subjects it falls to 0.534, the lowest of all; on the subject's own session it is not significantly different from FBCSP.

Figure 2. Mean accuracy by model and protocol (9 subjects; cross-session and LOSO (144) share test trials).

EEGNet Wins Every Held-out Subject



With 2,304 training trials EEGNet beats the Riemannian model in all nine held-out subjects (+0.095, 95% CI 0.055–0.136; pre-specified, Holm p = 0.012).

Figure 3. LOSO accuracy for each held-out subject (EEGNet: mean of 5 seeds).

Sensorimotor Signature

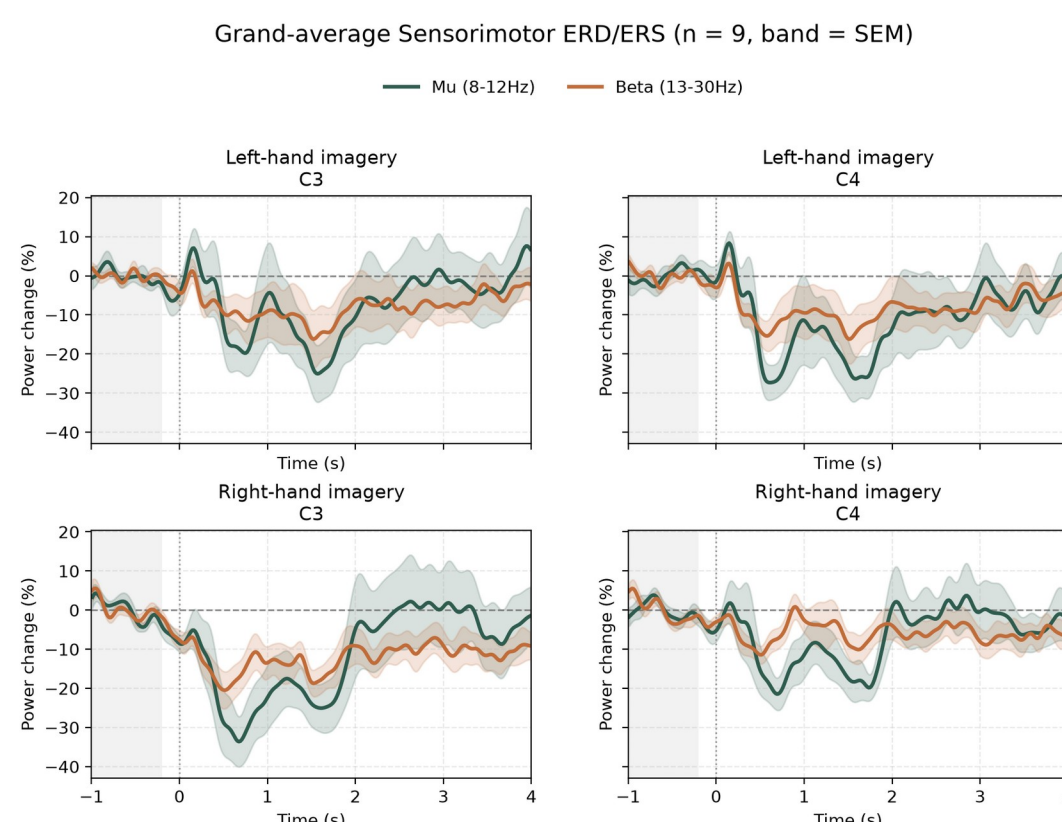
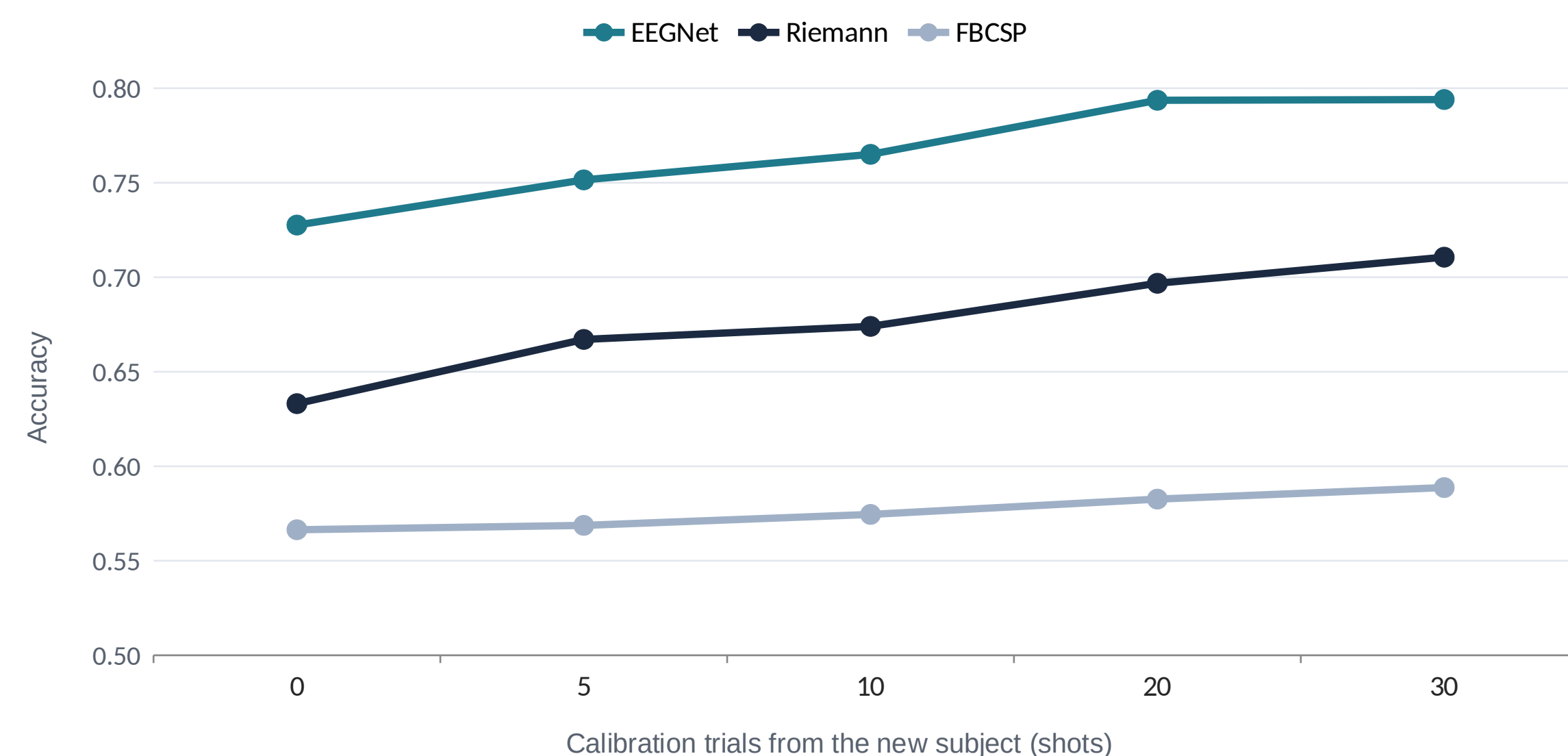


Figure 4. Grand-average ERD/ERS (n = 9, $\pm$ SEM). Mu and beta power fall after the cue at C3 and C4 (window means -6% to -18%). Contralateral dominance is significant only for right-hand beta (9/9 subjects, Holm p = 0.016).

Adapting to the Individual



Calibration lifts EEGNet from 0.728 to 0.794 and Riemann from 0.633 to 0.711; FBCSP barely moves. EEGNet beats Riemann in all 9 targets, but no test survives Holm correction.

Figure 5. Transfer accuracy against calibration trials per class (mean of 9 targets, 2 seeds).

Pre-specified Tests

Test (accuracy difference)	Difference [95% CI]	Holm p
P1 Cross-session: EEGNet – FBCSP	−0.055 [−0.130, 0.021]	0.20
P2 Cross-session: Riemann – FBCSP	−0.002 [−0.072, 0.069]	0.98
P3 LOSO: EEGNet – Riemann	+0.095 [0.055, 0.136]	0.012
A1 LOSO (144) vs cross: EEGNet – Riemann	−0.015 [−0.105, 0.076]	0.94
A2 LOSO (144) vs cross: EEGNet – FBCSP	+0.028 [−0.054, 0.111]	0.94

P1–P3 fixed before the runs; A1–A2 fixed in an addendum before the 144-trial runs. Exact sign-flip tests over 9 subjects; only P3 is significant.

Discussion

Data, not protocol. EEGNet's LOSO advantage vanishes when cross-subject training is cut to one session's worth of trials (A1, A2): it depends on a large pooled training set, not on cross-subject evaluation as such.

Own data counts. With the same 144 training trials, every model is 11–20 points more accurate when they come from the subject's own earlier session.

Evaluation decides rankings. Pooling sessions inflated within-subject accuracy by 2–6 points, and the corrected bug, normalisation and tests reversed several 2025 conclusions. Limits: 9 subjects, one dataset.

References

- BNCI Horizon 2020. 001-2014: Left and right hand motor imagery.
- Tangemann M, et al. Review of the BCI Competition IV. Front Neurosci 2012.
- Ramoser H, Müller-Gerking J, Pfurtscheller G. IEEE Trans Rehabil Eng 2000.
- Ang KK, et al. Filter bank common spatial pattern. Proc IJCNN 2008.
- Barachant A, et al. IEEE Trans Biomed Eng 2012.
- Lawhern VJ, et al. EEGNet. J Neural Eng 2018.
- Jayaram V, Barachant A. MOABB. J Neural Eng 2018.
- Gramfort A, et al. MNE software. NeuroImage 2013.
- Pfurtscheller G, Lopes da Silva FH. Clin Neurophysiol 1999.
- Holm S. Scand J Stat 1979.